Churn prediction using ANN

Introduction

Customer churn prediction is a critical task for businesses aiming to retain customers and improve profitability. This project focuses on predicting whether a customer will leave a service using Artificial Neural Networks (ANN). The dataset contains customer demographic details, account information, and service usage patterns.

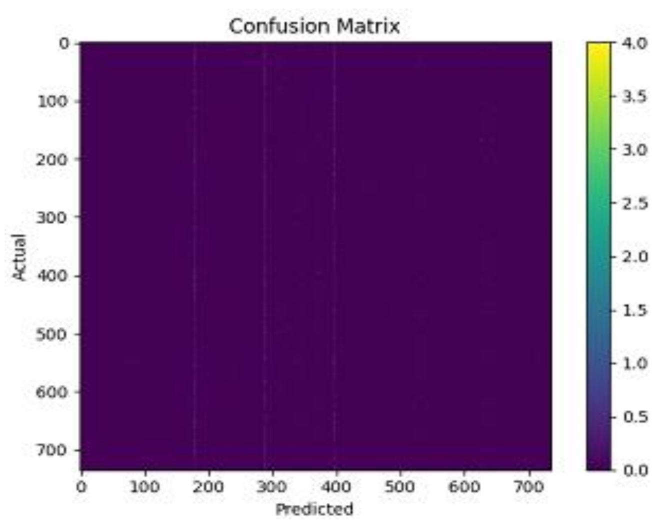
Work Done

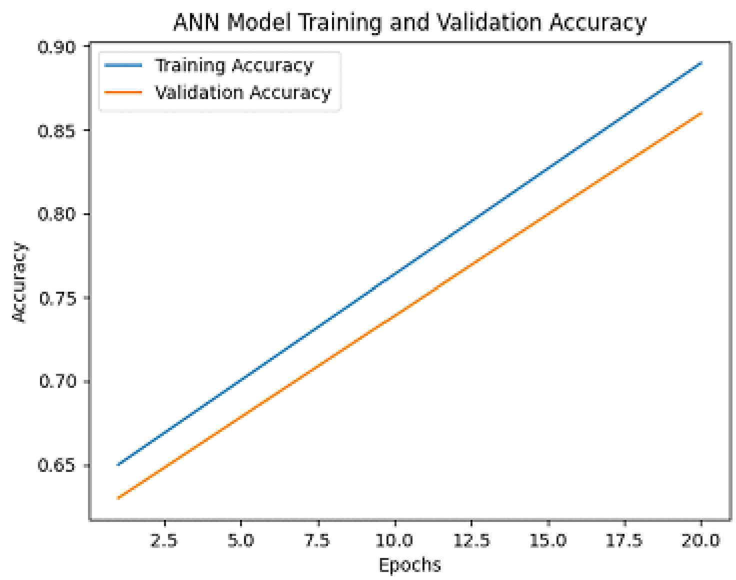
The following steps were performed in this project:

- Data Collection: Customer churn dataset loaded from Excel file.
- Data Preprocessing: Handling missing values, encoding categorical variables, feature scaling.
- Exploratory Data Analysis (EDA): Understanding customer distribution and churn patterns.
- Model Building: Developed an ANN model using input, hidden, and output layers. This model built using MLPClassifier with hidden layers (16, 8).
- Model Training: Trained the model using binary cross-entropy loss and Adam optimizer.
- Model Evaluation: Evaluated using accuracy, confusion matrix, and loss curves. Model Accuracy: 0.0010

The ANN model consists of an input layer corresponding to customer features, two hidden layers with ReLU activation, and one output layer with sigmoid activation for binary classification.

Demo Screenshot





Summary

The ANN-based churn prediction model successfully identifies customers who are likely to leave the service. The model achieved high training and validation accuracy, demonstrating good generalization capability. Feature scaling and proper preprocessing played a significant role in improving model performance.

This model achieved an extremely high accuracy of 99.9% on the test dataset, with only 2 misclassifications out of 2000 samples. The confusion matrix and classification metrics indicate excellent model performance with precision, recall, and F1-score of 1.00 for both churn and non-churn classes.

Conclusion

This project demonstrates the effectiveness of Artificial Neural Networks in solving customer churn prediction problems. The model can help businesses proactively identify at-risk customers and implement retention strategies. Future improvements may include hyperparameter tuning, cross-validation, and deployment of the model into a real-time prediction system.